

NONGBUALAMPHU, Thailand

WMO: 483600

Lat: 17.232N

Lon: 102.429E

Elev: 228

StdP: 98.61

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.4	15.3	7.1	6.4	18.0	9.1	7.4	18.9	5.9	21.5	5.2	22.0	0.8	N/A	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.5	38.5	24.7	37.0	24.7	35.7	24.5	27.2	32.3	27.0	31.9	26.7	31.3	2.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.3	28.8	25.9	21.9	28.6	25.7	21.5	28.4	88.2	32.6	86.7	32.0	85.5	31.5	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.8	4.1	3.4	10.6	39.8	2.1	1.7	9.1	41.0	7.8	42.0	6.6	43.0	5.1	44.3		
			DB	WB	8.8	28.5	1.8	0.9	7.5	29.1	6.5	29.6	5.5	30.1	4.2	30.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	22.9	25.6	28.1	30.1	29.5	29.1	28.3	27.9	27.8	27.1	25.9	23.2
	DBStd	3.00	2.63	2.51	2.70	2.50	1.76	1.37	1.54	1.26	1.15	1.35	1.78	2.68
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	6	3	0	1	0	0	0	0	0	0	0	0	2
	CDD10.0	6253	401	436	561	604	603	572	568	554	535	532	478	409
	CDD18.3	3218	146	203	304	354	345	322	309	296	285	273	228	153
	CDH23.3	34223	1171	2096	3604	4599	4036	3781	3353	2961	2827	2573	2003	1220
	CDH26.7	14431	357	968	1894	2553	1873	1606	1234	986	938	904	733	383
Wind	WSAvg	1.0	1.3	1.3	1.2	1.1	1.0	0.9	0.9	0.8	0.8	0.9	1.1	1.3
Precipitation	PrecAvg	1535	9	19	46	85	207	231	254	295	267	96	19	6
	PrecMax	1928	49	77	142	220	320	430	415	575	428	208	83	44
	PrecMin	1137	0	0	0	27	77	87	135	160	125	2	0	0
	PrecStd	211	15	19	36	45	66	67	71	91	85	56	25	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	36.6	39.6	41.1	38.6	36.5	36.7	34.0	34.2	33.7	34.1	33.1
		MCWB	21.9	22.4	23.1	25.7	26.3	25.9	25.6	26.5	26.4	25.2	24.6	23.1
	2%	DB	31.3	35.1	37.6	39.4	36.7	34.8	34.2	33.0	33.0	32.6	32.7	31.8
		MCWB	21.8	22.4	22.9	25.2	26.1	26.2	26.0	26.3	26.3	25.0	24.3	22.8
	5%	DB	30.1	33.6	36.0	37.6	35.2	33.7	32.8	32.0	32.1	31.9	31.6	30.2
		MCWB	21.6	22.2	23.1	25.2	26.0	26.2	26.0	26.1	26.1	24.7	23.7	22.2
	10%	DB	28.7	32.1	34.6	35.9	33.8	32.6	31.6	31.1	31.1	31.0	30.6	28.9
		MCWB	21.0	22.3	23.2	25.0	26.1	26.1	25.9	25.9	26.0	24.5	23.2	21.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.4	25.1	26.0	27.4	27.6	27.6	27.3	27.4	27.5	26.9	26.1	24.7
		MCDB	29.5	32.1	32.9	37.9	34.0	32.4	31.6	31.8	31.0	30.8	30.8	30.3
	2%	WB	23.4	24.1	25.2	26.4	27.1	27.1	26.8	26.9	27.0	26.3	25.4	23.9
		MCDB	28.4	31.2	32.1	35.4	33.4	31.9	30.9	30.9	31.1	30.0	30.1	28.9
	5%	WB	22.6	23.6	24.6	26.0	26.7	26.8	26.6	26.6	26.7	25.9	24.8	23.2
		MCDB	27.6	30.5	31.7	34.2	32.5	31.4	30.5	30.4	30.5	29.5	29.3	27.9
	10%	WB	21.9	23.0	24.2	25.5	26.4	26.5	26.2	26.2	26.4	25.5	24.2	22.3
		MCDB	26.8	29.7	31.3	33.2	31.8	30.9	30.0	29.8	29.9	28.9	28.6	27.5
Mean Daily Temperature Range	5% DB	MDBR	11.1	12.5	11.5	10.5	8.5	7.1	6.4	6.4	6.6	7.4	9.3	10.7
		MCDBR	12.7	14.8	14.5	12.8	10.6	8.8	8.5	7.7	8.0	8.8	10.4	11.9
		MCWBR	4.5	4.3	3.5	3.2	2.6	2.4	2.5	2.5	2.6	2.7	3.3	4.2
	5% WB	MCDBR	10.2	12.0	11.5	11.0	9.0	7.6	7.0	7.1	7.2	7.5	9.3	10.5
		MCWBR	4.1	4.2	3.5	3.4	2.9	2.6	2.6	2.7	2.8	2.7	3.4	4.1
Clear-Sky Solar Irradiance	taub	0.472	0.578	0.667	0.624	0.502	0.453	0.440	0.455	0.501	0.542	0.464	0.465	
	taud	2.089	1.832	1.665	1.808	2.171	2.334	2.367	2.308	2.164	2.006	2.203	2.130	
	Ebn at Noon	808	740	690	724	807	840	852	846	809	760	812	801	
	Edh at Noon	154	209	255	222	152	128	124	133	152	173	137	145	
All-Sky Solar Radiation	RadAvg	4.76	5.15	5.19	5.63	5.38	5.02	4.73	4.59	4.61	4.84	4.88	4.61	
	RadStd	0.31	0.25	0.29	0.24	0.33	0.33	0.29	0.29	0.28	0.33	0.18	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+119

Nomenclature: See separate page